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Natural Language Processing - Probability Models in Python

Today's goal • 3/3 items

Module 1
Welcome

- Introduction to the Course 'Natural Language Processing - Probability Models in Python'
  Reading • 10 min
- Introduction and Outline
  Video • 5 min
- Full Course Resources
  Reading • 10 min
- Special Offer
  Video • 1 min
- Where to Get the Code
  Video • 2 min

Module 2
Markov Models

Module 3
Article Spinner

Module 4
Cipher Decryption

Introduction to the Course 'Natural Language Processing - Probability Models in Python'

0:06 / 2:39 2x

Welcome to the course **Natural Language Processing - Probability Models in Python**. The Second of the 3 course Specialization in Modern Natural Language Processing

The course is structured into Four comprehensive modules:

1. **Welcome:** In this module, we will introduce the course, providing an overview of the key topics and concepts to be covered. You'll also learn how to access important resources, such as special offers and the course code, to enhance your learning experience and ensure you have everything needed to get started.
2. **Markov Models:** In this module, we will explore the fundamentals of Markov models and their application in Natural Language Processing. You'll learn how to build probabilistic text classifiers and language models by understanding state transitions, applying smoothing techniques, and coding real-world NLP solutions in Python. By the end of the section, you'll have implemented your own models to classify and generate text based on probability-driven methods.
3. **Article Spinner:** In this module, we will delve into the concept of article spinning and how to generate diverse and unique content. We'll explore the n-gram approach for text variation, code an article spinner in Python, and discuss real-world issues in spinning content. By the end, you'll be able to create functional and meaningful article spinners that produce varied text while avoiding common mistakes.
4. **Cipher Decryption:** In this module, we will explore the use of probability models in cipher decryption, focusing on genetic algorithms and language models. You'll learn how to implement and optimize decryption algorithms in Python to crack encrypted messages. Additionally, we'll explore real-world applications like acoustic keyloggers and discuss the significance of decryption in maintaining digital security.

Target Learner: The course is perfect for learners who want to expand their skills in NLP using probability models, from beginners looking to understand the basics to intermediate learners seeking to enhance their coding expertise. You will gain the confidence and skills to apply these models to solve real-world NLP problems such as text classification and decryption.

Learning Objectives:

1. Build and implement Markov models for text classification tasks in Python.
2. Create and refine language models to predict word sequences and improve text generation.
3. Understand and apply probability smoothing techniques to handle rare events in NLP tasks.
4. Use genetic algorithms in cipher decryption to crack encrypted messages using probability models.

Go to next item ✓ Completed

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